Consumer Surplus & Producer Surplus

Based on Ch. 5, Economics 10th ed., Parkin, Powell & Mathews

Course: Eco 101 (Introduction to Microeconomics)

Consumers' Surplus

Equilibrium can be explained using two key ideas : ***consumers' surplus*** and ***producers' (or sellers') surplus***

❑ Consumers' surplus (*CS*):

What is it?
The extra benefit a buyer gets when they pay **less than what they were willing to pay**.

Formula:
Consumer Surplus = Maximum buying price – Actual Price Paid

Example:
If you're willing to pay **$10** for a movie ticket, but you only pay **$7**,
➤ Your **Consumer Surplus = $3**

Why it matters:
The **more surplus** consumers get, the **better off** they are.

Consumer surplus shows the **value or benefit** buyers get from being in the market.

Consumer Surplus on a Graph

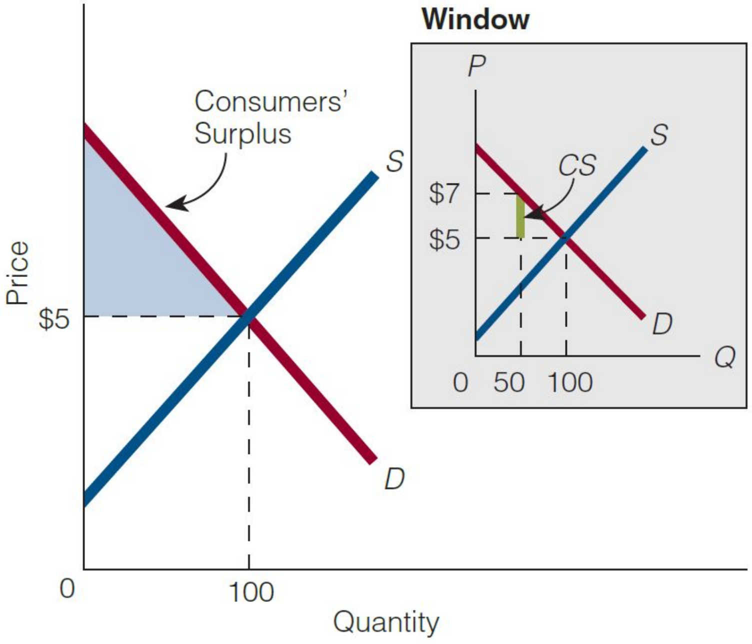
Visually:
-Consumer Surplus is the area **under the demand curve** and **above the equilibrium price line**.

What Does the Graph Show?

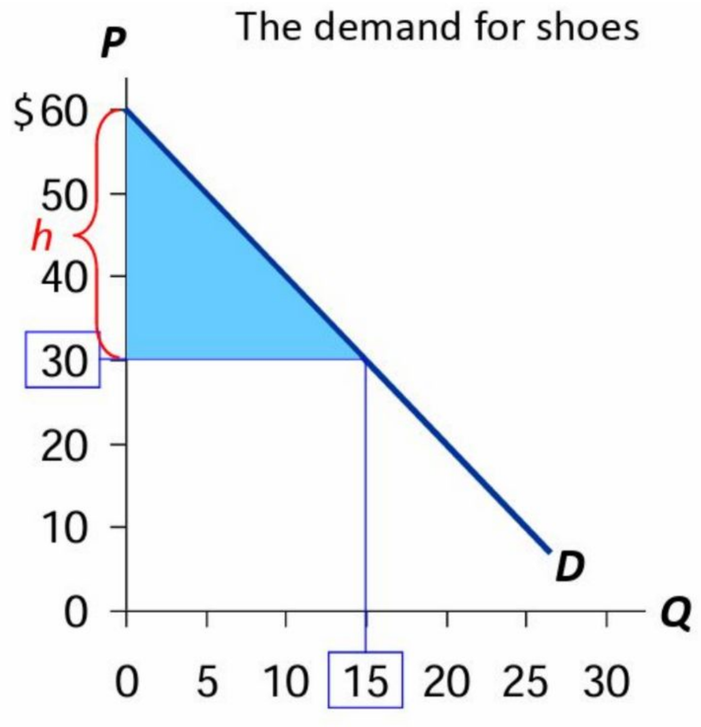
- At the **50th unit**, buyers are **willing to pay $7**, but they **only pay $5**
  ➤ This creates a **$2 surplus** on that unit.
- This happens for **every unit** from the **1st to the 100th**.

Total Consumer Surplus

- Add up all the surpluses from each unit...
  ➤ It forms a **shaded triangle** on the graph.
- At a price of **$5**,
  ➤ **Consumer Surplus = Area of the shaded triangle**



Consumer Surplus – Example



- Area of triangle= ½ x base x height
- Height of triangle = 60 - 30 = 30
- Base of triangle = 15
- **CS** = ½ x 30 x 15 = **$225**

Producers' Surplus

❑ Producers' surplus (*CS*):

What is it?
The extra benefit a seller gets when they **receive more** than the **minimum they were willing to accept**.

Formula:

Producer Surplus = Price Received – Minimum Selling Price

🎬 **Example:** The movie theater owner is willing to accept **$5** per ticket, but sells it for **$7**
➢ Her **Producer Surplus = $2**

💡 Why it Matters:

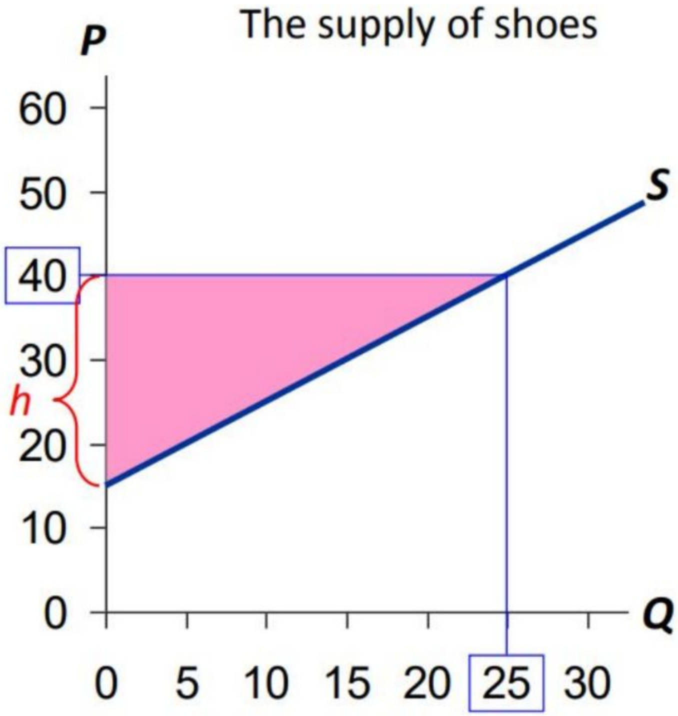
- The **more surplus** producers receive, the **better off** they are.
- Producer surplus shows the **benefit sellers get** from taking part in the market.

Producer Surplus– Example

PS is the area b/w ***P*** and the ***S*** curve, from 0 to ***Q***.

The height of this triangle is $40 – 15 = $25.

So,
PS = ½ x *b* x *h*
= ½ x 25 x $25
= <u>$312.5</u>



📉 Producer Surplus on a Graph

Visually:

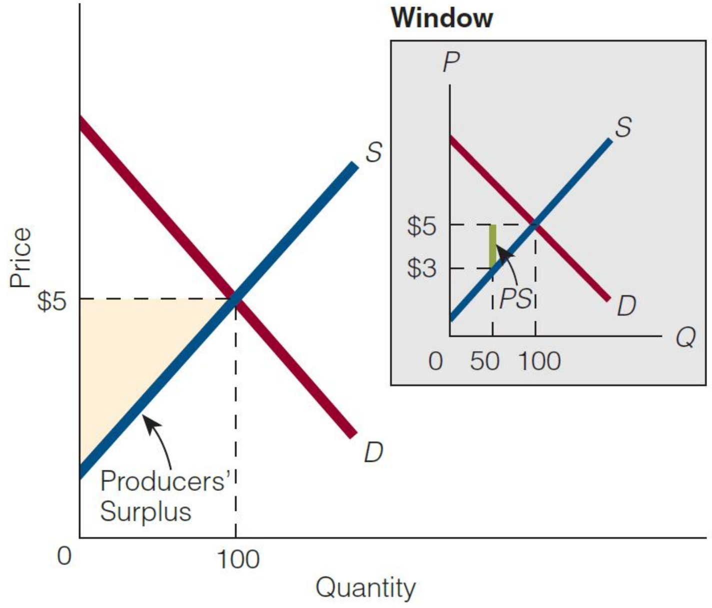
- Producer Surplus is the area **above the supply curve** and **below the equilibrium price line**.

What Does the Graph Show?

- For the **50th unit**, sellers were willing to sell it for **$3**, but actually sold it for **$5**
  ➢ This creates a **$2 surplus** on that unit.
- This happens for **each unit** from the **1st to the 100th**.

🔺 Total Producer Surplus

- When we **add up all the surpluses**,
  ➢ It forms a **shaded triangle** on the graph.
- At a price of **$5**,
  ➢ **Producer Surplus = Area of the shaded triangle**



Total Surplus

Total surplus (*TS*) is the sum of consumers' surplus and producers' surplus:

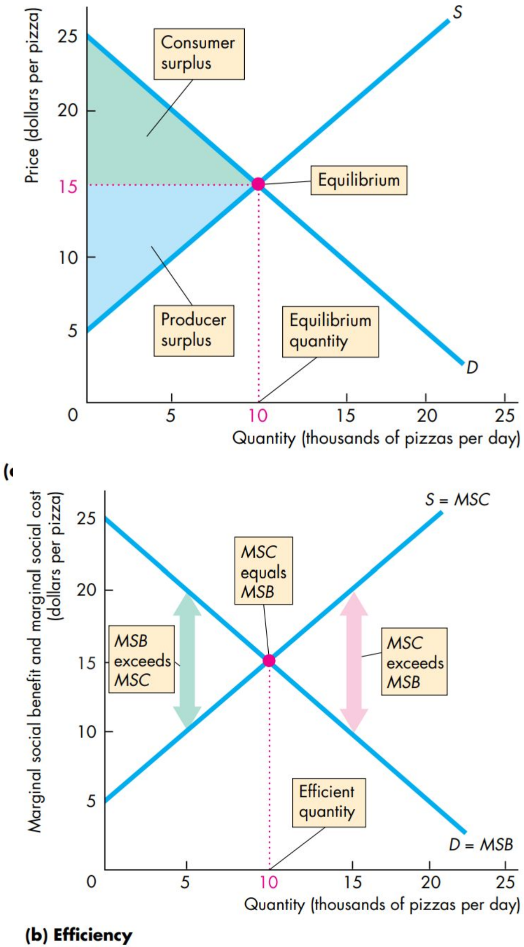
Total surplus = Consumers' surplus + Producers' surplus

⭐ Why Equilibrium Matters

At **equilibrium**, both buyers and sellers are doing the best they can.

- ➢ **Consumer Surplus is maximized**
- ➢ **Producer Surplus is maximized**
- ➢ So, **Total Surplus is maximized**

FIGURE 5.5 An Efficient Market for Pizza



Equilibrium and Surpluses

Figure 5.5(a) also shows the consumer surplus and producer surplus. The sum of consumer surplus and producer surplus is called total surplus. When the efficient quantity is produced, total surplus is maximized. Buyers and sellers acting in their self-interest end up promoting the social interest.